Chapter 16 Review

Campbell Biology, AP Edition, 12e

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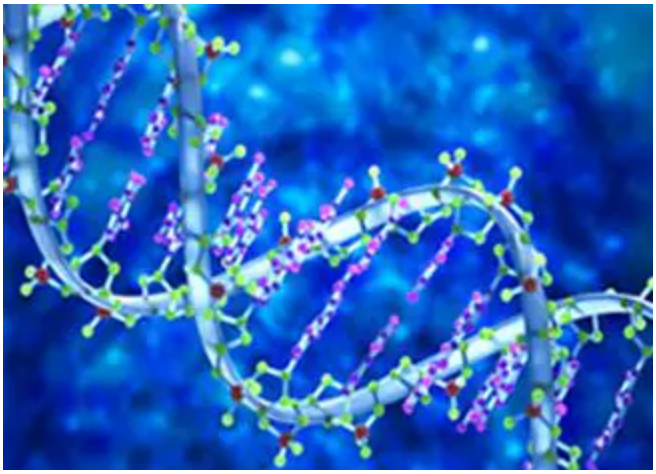
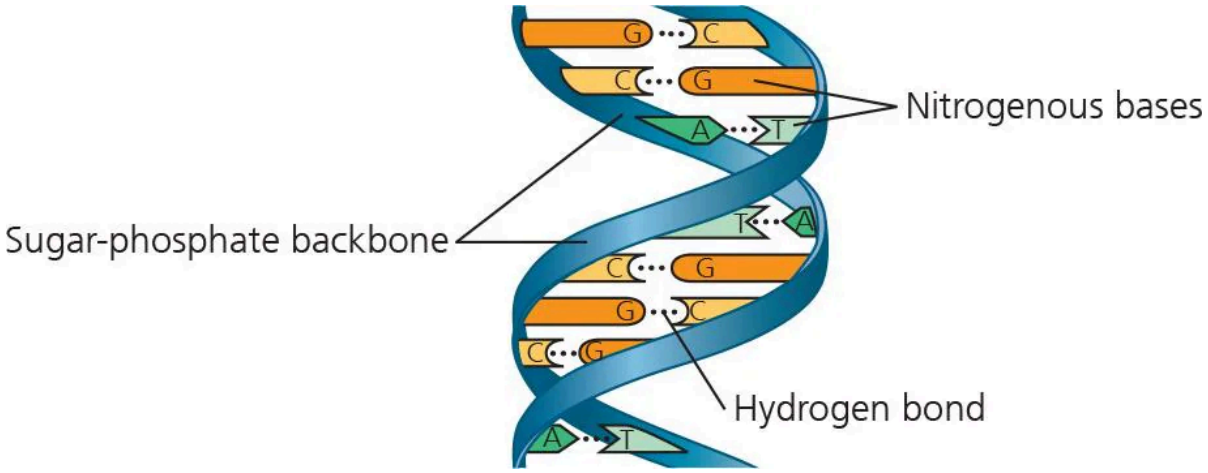


Summary of Key Concepts

Hershey and Chase concluded that DNA is the source of heritable information. Select the data from their experiments that support that claim. **(BIG IDEA 3)**

Concept 16.1: DNA is the genetic material

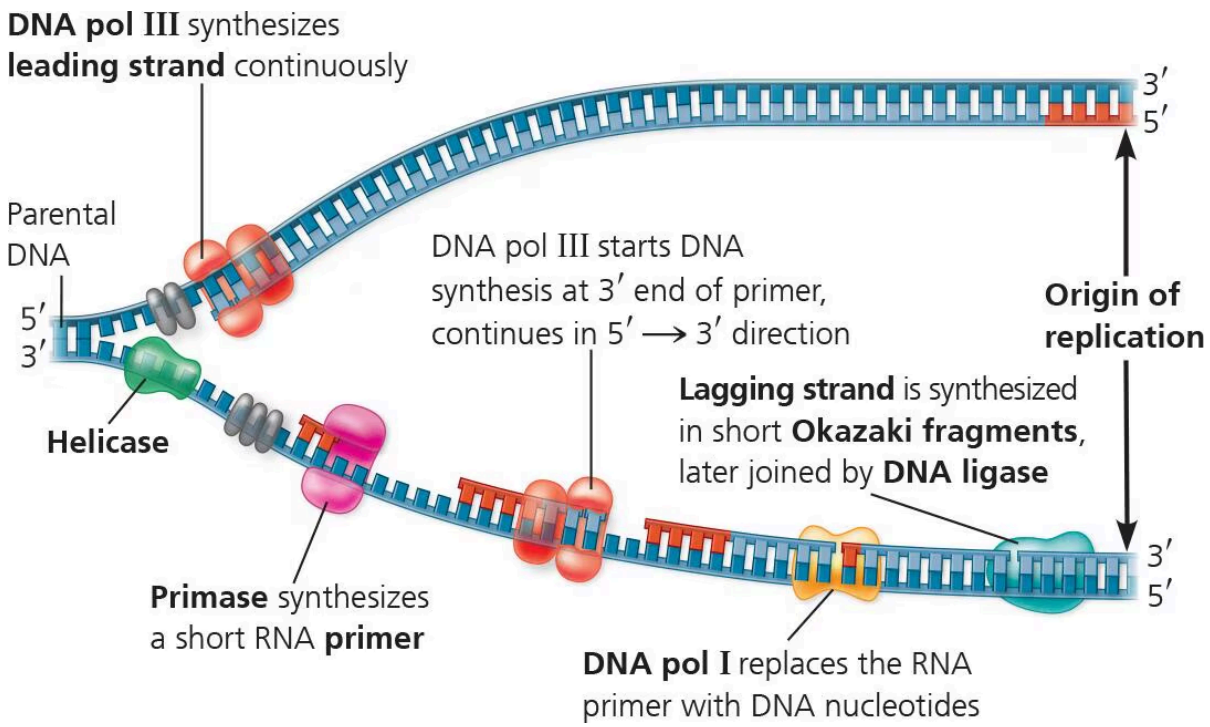
- Experiments with bacteria and with **phages** provided the first strong evidence that the genetic material is DNA.
- Watson and Crick deduced that DNA is a **double helix** and built a structural model. Two ****__antiparallel__ sugar-phosphate chains wind around the outside of the molecule; the nitrogenous bases project into the interior, where they hydrogen-bond in specific pairs, A with T, G with C.



What does it mean when we say that the two DNA strands in the double helix are antiparallel? What would an end of the double helix look like if the strands were parallel?

Concept 16.2: Many proteins work together in DNA replication and repair

- The Meselson-Stahl experiment showed that **DNA replication** is **semiconservative**: The parental molecule unwinds, and each strand then serves as a template for the synthesis of a new strand according to base-pairing rules.
- DNA replication at one **replication fork** is summarized here:



Left side replication fork shown. Top parental D N A strand runs (left to right) 5 prime to 3 prime. Bottom strand runs 3 prime to 5 prime. D N A pol 3 synthesizes leading (top template) strand continuously from the origin of replication (point near the middle of the replication bubble on both the top and bottom template strands) towards the replication fork. RNA primer is drawn for a few base lengths at the origin. Sliding clamp pushes D N A pol 3 towards the replication fork. Single strand binding proteins and helicase also on parental template near the fork. In this diagram, the bottom parental strand (3 prime at the left and 5 prime at the right) is the template for the lagging strand.

1. Primase synthesizes short strands of R N A primer.
2. D N A pol 3 starts D N A synthesis at the 3 prime end of R N A primer and continues in the 5 prime to 3 prime direction.
3. D N A pol 1 replaces RNA primer between lagging strand fragments with D N A nucleotides.
4. Lagging strand is synthesized in short Okazaki fragments, later joined by D N A ligase.

- ****DNA polymerases** proofread new DNA, replacing incorrect nucleotides. In ****mismatch repair**, enzymes correct errors that persist. ****Nucleotide excision repair** is a process by which ****nucleases** cut out and other enzymes replace damaged stretches of DNA.
- The ends of eukaryotic chromosomal DNA get shorter with each round of replication. The presence of ****telomeres**, repetitive sequences at the ends of linear DNA molecules, postpones the erosion of genes. Telomerase catalyzes the lengthening of telomeres in germ cells.

Compare DNA replication on the leading and lagging strands, including similarities and differences.

Concept 16.3: A chromosome consists of a DNA molecule packed together with proteins

- The chromosome of most bacterial species is a circular DNA molecule with some associated proteins, making up the nucleoid.
- The ****__chromatin__ making up a eukaryotic chromosome is composed of DNA, **histones**, and other proteins. The histones bind to each other and to the DNA to form **nucleosomes**, the most basic units of DNA packing, which exist as part of the 10-nm fiber. Histone tails extend outward from each bead-like nucleosome core. Additional coiling and folding lead ultimately to the highly condensed chromatin of the metaphase chromosome.
- Chromosomes occupy restricted areas in the interphase nucleus. In interphase cells, most 10-nm fiber chromatin is loosely arranged ****(**euchromatin**), but some is more densely arranged (**heterochromatin**). Euchromatin, but not heterochromatin, is generally accessible for gene transcription.

Describe the levels of chromatin packing you'd expect to see in an interphase nucleus.